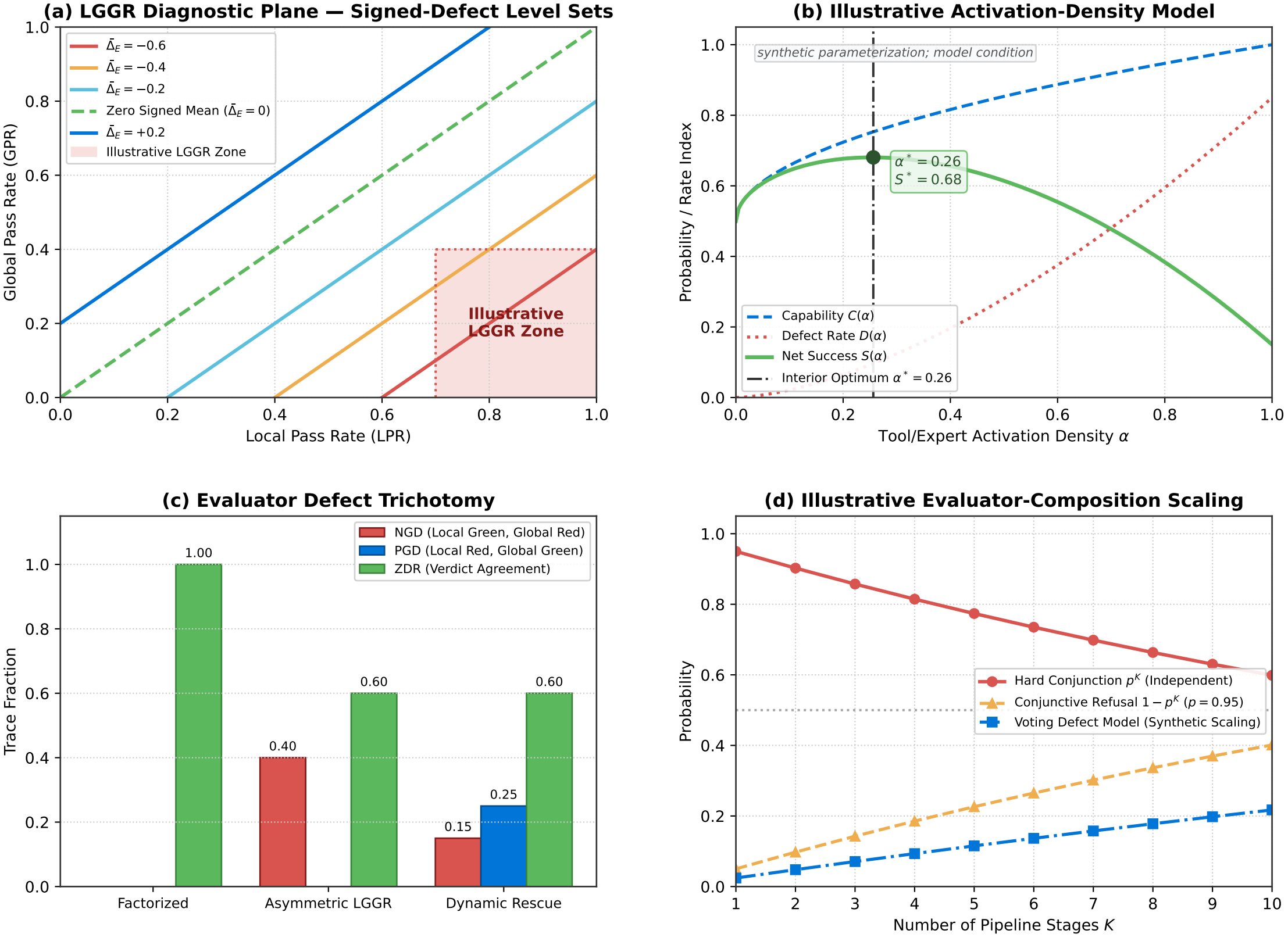
(a) LGGR Diagnostic Plane — Signed-Defect Level Sets
$\bar{\Delta}_E = -0.6$
$\bar{\Delta}_E = -0.4$
$\bar{\Delta}_E = -0.2$
Zero Signed Mean ($\bar{\Delta}_E = 0$)
$\bar{\Delta}_E = +0.2$
Illustrative LGGR Zone
Illustrative LGGR Zone
Global Pass Rate (GPR)
Local Pass Rate (LPR)
(b) Illustrative Activation-Density Model
synthetic parameterization; model condition
$\alpha^* = 0.26$
$S^* = 0.68$
Capability $C(\alpha)$
Defect Rate $D(\alpha)$
Net Success $S(\alpha)$
Interior Optimum $\alpha^* = 0.26$
Probability / Rate Index
Tool/Expert Activation Density α
(c) Evaluator Defect Trichotomy
NGD (Local Green, Global Red)
PGD (Local Red, Global Green)
ZDR (Verdict Agreement)
1.00
0.40
0.60
0.15
0.25
0.60
Trace Fraction
Factorized
Asymmetric LGGR
Dynamic Rescue
(d) Illustrative Evaluator-Composition Scaling
Hard Conjunction p^K (Independent)
Conjunctive Refusal $1 - p^K$ ($p = 0.95$)
Voting Defect Model (Synthetic Scaling)
Probability
Number of Pipeline Stages K